

Strategic Dossier: Consumer Claims Agent for Protex Hack-2-Win 2026

SECTION 1: Executive conclusion

The Consumer Claims Agent represents a highly strategic, high-impact product concept designed explicitly to dominate Track 2 (Agentic AI / AI Workflows) at the Protex Hack-2-Win Hackathon 2026. This product functions as an autonomous grievance-resolution copilot, engineered to bridge the profound asymmetry between frustrated Indian consumers and the complex, opaque policies of massive corporations. The core functionality centers on ingesting unstructured user evidence—such as screenshots of receipts, damaged product photos, and brief complaint descriptions—and autonomously cross-referencing this data against official corporate policies and government regulations. The system then executes a multi-step reasoning protocol to determine the validity of the claim and autonomously generates a structured, legally sound escalation package directed at the appropriate corporate nodal officer.

For a team of four beginner developers operating within a constrained eight-day build window, this concept offers an optimal balance of technical feasibility and demonstrative spectacle. The architecture leverages visual orchestration platforms, specifically n8n, combined with robust API integrations for optical character recognition and large language model reasoning, entirely circumventing the need for complex, ground-up machine learning model training. This perfectly aligns with an AI-assisted, low-code-friendly development methodology while delivering a product that feels exceptionally sophisticated.

The analysis indicates three primary reasons this product will heavily impress the judging panel. First, the product allows for highly visible analytical processing. By designing an "Agentic User Interface" that streams the system's internal reasoning, tool-calling, and policy comparison steps in real-time, the project visually proves its autonomy rather than acting as a black-box text generator. Second, the concept demonstrates extraordinary real-world utility and market urgency. The project addresses a systemic, heavily documented Indian market failure, capitalizing on recent government pushes for digital justice and surging national grievance statistics. Third, the product exhibits strict adherence to the Track 2 criteria. It transcends standard chatbot functionality by operating via a "Plan-and-Execute" architecture, utilizing external tools to retrieve data and construct complex documents without requiring constant, granular human prompting.

Conversely, the implementation strategy must account for three critical risks. First, the fragility of Optical Character Recognition (OCR) presents a significant point of failure; poorly lit or crumpled receipt images could derail the data extraction phase during a live demonstration. Second, the potential for regulatory hallucination remains high; without strict grounding in a

vector database, the language model might invent non-existent consumer protection laws, instantly destroying the product's credibility. Third, there is a risk of Unauthorized Practice of Law (UPL) perception. If the product is marketed or framed as an "AI Lawyer" rather than an administrative drafting assistant, it will attract severe scrutiny regarding legal liability, ethical compliance, and user safety.

SECTION 2: Why this idea matters

The consumer grievance redressal framework in India is currently buckling under unprecedented strain, rendering accessible, pre-litigation resolution tools an absolute necessity. Recent data from the Consumer Justice Report 2026 reveals a sharp 21% surge in case pendency across consumer commissions between 2020 and 2024, with the national backlog exceeding 5.15 lakh cases.¹ This bottleneck means statutory resolution timelines of three to five months are routinely violated, undermining consumer confidence in formal judicial remedies.¹ Concurrently, alternative resolution platforms like the National Consumer Helpline (NCH) have experienced exponential growth, with call volumes skyrocketing from 12,553 in December 2015 to over 1,55,138 by December 2024, and the platform now processing over one lakh complaints monthly.²

This friction is most heavily concentrated within digital and service-based transactions. The e-commerce sector consistently dominates the grievance landscape; in July 2025 alone, the NCH facilitated e-commerce refunds amounting to ₹1.34 crore across thousands of disputes.² Similarly, the Indian aviation sector frequently generates massive consumer distress. For instance, the mass flight cancellations by IndiGo in late 2025, triggered by crew shortages and updated Directorate General of Civil Aviation (DGCA) flight duty rules, left thousands of passengers stranded.⁵ While DGCA regulations stipulate clear monetary compensation for such cancellations, airlines often default to offering less valuable travel vouchers unless the consumer explicitly cites the law and demands cash restitution.⁶

The current consumer experience is defined by severe administrative and emotional friction. When an issue arises, the consumer must first aggregate scattered evidence across emails, application screenshots, and physical invoices. Next, they face immense policy opacity; corporate refund rules are often buried within dense terms of service, and relevant government protections—such as the Consumer Protection (E-Commerce) Rules 2020 or the DGCA Passenger Charter—are written in complex legal vernacular.⁹ Furthermore, finding the correct escalation path is intentionally difficult, with nodal officer contact details frequently hidden deep within corporate registries.¹¹ The cumulative effect of these barriers is an exceptionally low follow-through rate. Consumers routinely abandon valid claims worth thousands of rupees because the time cost, cognitive load, and emotional frustration required to fight the corporate bureaucracy simply outweigh the financial recovery.

This problem is particularly acute and relevant in 2026. The Indian government is currently executing a massive push toward digital consumer justice, evidenced by the rollout of the

e-Jagriti platform, which digitized over 1.31 lakh consumer cases in 2025.¹³ Furthermore, regulatory bodies like the Central Consumer Protection Authority (CCPA) are actively cracking down on deceptive digital designs through comprehensive dark pattern guidelines.¹⁵ Within this precise macro-economic context, an agentic AI workflow is vastly superior to a simple chatbot. A conversational chatbot places the cognitive burden on the user, requiring them to know exactly which questions to ask and which laws to reference. Conversely, an agentic workflow is goal-oriented. It takes raw, unstructured inputs, evaluates the context, independently retrieves the relevant legal parameters, and executes the administrative labor required to build a compelling case, fully aligning with the current national mandate for accessible digital justice.

SECTION 3: Track fit analysis

The Consumer Claims Agent aligns flawlessly with the mandates of Track 2: Agentic AI / AI Workflows. The official framing of this track demands the construction of autonomous agents capable of multi-step reasoning, tool use, and real-world task execution. The proposed product does not merely simulate these requirements; its core architecture depends entirely on them.

The product maps explicitly to the concept of autonomous behavior. Once a user uploads a receipt and types a brief description of their grievance, the system assumes complete control of the execution flow. It does not require a conversational back-and-forth to proceed; instead, it dictates the sequence of operations, moving from data extraction to policy retrieval to document generation without intermediate human prompting. This is the defining characteristic of an autonomous workflow.

Furthermore, the system demonstrates deep multi-step reasoning through the ReAct (Reason + Act) design pattern.¹⁷ The agent must first deduce the vendor and the transaction date from an unstructured image. It must then reason about the category of the complaint to determine which vector database to query. The most complex reasoning step occurs when the agent cross-references the retrieved corporate policy against the retrieved government regulation to mathematically and logically determine if the user has a valid claim for compensation.⁸

Tool use is heavily integrated into the architecture. The agent cannot achieve its goal relying solely on its internal neural weights. It must invoke an external Optical Character Recognition (OCR) API to parse the uploaded image, it must trigger a vector database search to retrieve current regulatory text, and it must utilize a document generation tool to format the final output.²⁰ Finally, the system executes a real-world task by producing a formatted, legally cited escalation email directed to a specific, real-world corporate nodal officer, transforming a digital interaction into an actionable administrative artifact.¹¹

To ensure the project feels truly agentic to the judging panel, the implementation must aggressively avoid "black box" processing. A system that simply displays a static loading spinner for thirty seconds before producing a final email feels indistinguishable from a

standard, heavily scripted API call. To feel genuinely agentic, the user interface must expose the agent's internal state. By utilizing an interface that streams the system's thought process—displaying real-time updates such as *"Extracting PNR from image..."*, *"Querying DGCA CAR database for delay compensation tiers..."*, and *"Evaluating vendor compliance..."*—the team will visually prove the existence of an autonomous orchestration loop.²¹ This transparency is what separates a winning agentic submission from a generic generative text wrapper.

SECTION 4: User and market analysis

The primary user of the Consumer Claims Agent is the digitally active, middle-income Indian consumer. This demographic conducts frequent online transactions across e-commerce, mobility, and aviation platforms but generally lacks the legal literacy, time, or financial resources to hire professional counsel for mid-ticket disputes (ranging from ₹2,000 to ₹25,000). The secondary users include small and medium enterprise (SME) owners who frequently manage vendor disputes and procurement refunds but lack dedicated legal departments. The ideal initial user segment for product launch comprises urban professionals aged 25-45, who value their time highly and experience a high volume of digital transaction disruptions.

The core Job-To-Be-Done (JTBD) for this user is clear: "When a service provider fails to deliver, help me secure my rightful refund or compensation without forcing me to spend hours researching opaque policies and fighting automated customer support bots." The exact moment of need occurs immediately after a primary customer support channel fails—for example, when an airline sends a generic email offering a travel voucher instead of a mandated cash refund, or when an e-commerce platform automatically rejects a return request for a damaged electronic device.²²

To maximize demonstration impact and align with beginner development capabilities, the hackathon MVP must be restricted to highly specific complaint verticals. Attempting to build a generalized legal agent will result in severe hallucination and demo failure.

Complaint Category	Demo Impact	Data Availability	Legal / Policy Clarity	Beginner Feasibility	Emotional Resonance
Aviation (Delays/Cancellations)	Very High	High (Standardized tickets, PNRs) ²⁴	Very High (DGCA rules dictate exact monetary tiers) ⁶	High	Very High

E-commerce (Damaged/ Wrong Item)	High	High (Standardized invoices) ²⁵	High (CPA 2019, CCPA Rules) ⁹	High	High
Telecom (Billing Errors)	Medium	Medium (Complex, multi-page bills)	Medium (TRAI regulations are dense and hard to parse)	Low	Medium
SaaS Subscriptions	Low	Low (Hard to prove cancellation attempts visually)	Low (Subject to varied international jurisdictions)	High	Low
Financial (Failed UPI/Refunds)	High	Low (Banking data is highly secure and format-varied)	High (RBI turnaround time guidelines)	Low	Very High

Based on this analysis, the recommended initial complaint verticals for the hackathon build are **Aviation** and **E-commerce**. Aviation provides the absolute strongest demonstration scenario because the DGCA Passenger Charter contains strict, mathematical compensation metrics (e.g., exactly ₹10,000 for specific cancellation windows) that an AI agent can logically deduce and apply to a specific ticket.⁶ E-commerce provides a highly relatable secondary scenario, allowing the agent to demonstrate conflict resolution between a platform's restrictive internal return policy and the overriding national Consumer Protection (E-Commerce) Rules 2020.²²

SECTION 5: Competitive landscape

The current market features a variety of overlapping solutions regarding consumer grievances and legal assistance, yet a significant whitespace exists for an autonomous, consumer-facing

administrative copilot in the Indian market.

The primary incumbent category consists of official government complaint portals, such as the e-Jagriti platform and the National Consumer Helpline (NCH).¹⁴ These platforms are highly authoritative and effectively facilitate alternative dispute resolution. However, they represent the destination of a complaint. They do not assist the user in the preliminary stages of analyzing personal evidence, comparing it against corporate policies, or drafting the initial escalation to the company itself. They require the user to arrive with a fully formed, well-articulated grievance.

The second category encompasses traditional legal-tech tools, such as CaseMine.²⁸ While these platforms possess powerful AI retrieval capabilities, their target users are practicing lawyers and enterprise legal departments. They excel at retrieving complex judicial precedents and case law visualization but are completely unsuited, both in user experience and pricing, for a retail consumer seeking to resolve a ₹5,000 flight cancellation.

The third category includes consumer advocacy startups, globally pioneered by companies like DoNotPay.²⁸ DoNotPay initially excelled at automating bureaucratic tasks like fighting parking tickets. However, their strategic positioning serves as a critical cautionary tale. By aggressively marketing themselves as the "world's first robot lawyer" and claiming to replace human legal expertise, they attracted severe regulatory backlash, culminating in Federal Trade Commission (FTC) enforcement actions in 2024 for deceptive AI claims and the unauthorized practice of law.²⁹

The final category involves DIY complaint drafting tools, such as LegalZoom or LawDepot.³⁰ These platforms offer static legal templates. While they avoid unauthorized practice of law liabilities, they place the entire cognitive burden on the user, requiring them to manually parse their own evidence and understand exactly which template applies to their situation. They lack any autonomous evaluation or reasoning capabilities.

Summary of White Space and Positioning:

The Consumer Claims Agent occupies the lucrative white space between static DIY templates and high-risk "robot lawyers." It differentiates itself by acting purely as an administrative data processor and escalation structured. It does not file lawsuits, it does not provide legal advice, and it does not guarantee outcomes. By explicitly focusing on pre-litigation corporate escalations—drafting customer service emails based on publicly available policies—the product mitigates legal risk while delivering immense value. The judging panel will likely react highly favorably to this positioning, as it demonstrates not only technical execution but a mature, grounded understanding of regulatory realities and product safety.

SECTION 6: Product definition

One-line concept: An autonomous claims copilot that transforms scattered consumer

evidence into structured, policy-backed corporate grievance escalations.

30-second explanation: When a major airline cancels a flight or an e-commerce platform denies a valid return, consumers are forced to navigate complex terms of service and draft tedious escalation emails to customer support. The Consumer Claims Agent automates this friction. It takes a screenshot of the user's receipt and a brief description of the problem, autonomously researches the specific corporate policies and overriding consumer laws, and generates a complete, legally sound escalation package ready to be sent directly to the company's designated nodal officer.

Clear user promise: Reclaim your money and your time without needing to read a single page of legal jargon.

Product boundaries and what the product is not: The product is strictly an administrative structuring and research tool. It is absolutely not a legal representative. It does not file formal complaints with consumer courts, it does not negotiate directly with corporate entities on the user's behalf, it does not establish attorney-client privilege, and it does not guarantee financial recovery. It is a document preparation assistant.

Exact inputs, outputs, and user value:

- **Inputs:** A natural language description of the complaint (e.g., "They delivered a broken TV and won't take it back") and uploaded visual evidence (PDF invoices, PNG screenshots of flight tickets).
- **Outputs:** A comprehensive Case Summary outlining the extracted facts, a Policy Context Brief citing the specific rules that have been violated, and a formatted Escalation Draft addressed to the appropriate nodal officer.
- **User Value:** The drastic reduction of cognitive load and time expenditure, coupled with a higher probability of corporate compliance due to the professional, policy-cited nature of the escalation.

Framing Tradeoffs: Framing the product as a "Personal Consumer Advocate" sounds highly appealing to retail users but implies a fiduciary duty and ongoing representation that the software cannot fulfill. Framing it as a "Legal-Tech Assistant" sounds sophisticated to investors but immediately triggers Unauthorized Practice of Law (UPL) scrutiny and invites judgment against the rigorous standards of professional legal software.³¹

The recommended approach is to frame the product strictly as a "**Grievance-Resolution Copilot.**" This semantic choice implies assistance, collaboration, and administrative support rather than legal authority. It anchors the product in the realm of customer service and corporate accountability rather than judicial litigation, drastically lowering the liability profile while maintaining a high perceived value for the hackathon presentation.

SECTION 7: Deep agent workflow design

To build a system that moves beyond linear processing into true autonomy, the workflow must utilize a "Plan-and-Execute" or "Router-Specialist" architectural pattern.³² The system must be capable of deciding which specific tools to call based on the evaluation of preceding steps. The detailed workflow is structured into three primary phases: Intake & Extraction, Cognitive Processing, and Execution.

Phase 1: Intake & Extraction

1. Complaint Intake

- **Trigger & Objective:** User submits a web form. The objective is to capture the baseline narrative and intent.
- **Tool / Mechanism:** React UI triggering an n8n webhook.
- **Input / Output:** Raw text -> JSON payload.
- **Decision Logic:** Deterministic. If the text is below a minimum character count, the system halts and prompts for more detail.

2. Evidence Processing

- **Trigger & Objective:** Triggered simultaneously with step 1. The objective is to convert visual data into machine-readable text.
- **Tool / Mechanism:** AWS Textract (Analyze Document API).²⁰
- **Input / Output:** PDF/JPG -> Raw extracted text blocks and key-value pairs.
- **Failure Mode / Fallback:** If the image is illegible and OCR confidence is below a threshold, the agent falls back to conversational mode, requesting the user manually input their PNR or Invoice Number.

3. Entity Extraction & Timeline Reconstruction

- **Trigger & Objective:** Triggered by successful OCR. The objective is to identify the core facts.
- **Tool / Mechanism:** Large Language Model (Prompt-only, e.g., GPT-4o-mini).
- **Input / Output:** Raw OCR text + User Complaint -> Structured JSON (Vendor Name, Transaction Date, Amount, Item, Order ID).
- **Decision Logic:** Probabilistic. If the LLM cannot confidently identify a vendor, the agent pauses execution and requests human approval/clarification.

Phase 2: Cognitive Processing

4. Complaint-Type Classification

- **Trigger & Objective:** Triggered by extracted entities. The objective is to route the agent to the correct domain logic.
- **Tool / Mechanism:** LLM Classification prompt.
- **Input / Output:** Vendor + Item -> Categorical tag (e.g., "Aviation_Cancellation", "Ecom_Damaged").

- **Autonomy Element:** This step creates autonomy. The agent dynamically alters its subsequent execution pathway based purely on this tag, acting as an intent router.³⁴

5. Policy Retrieval

- **Trigger & Objective:** Triggered by classification. The objective is to find the specific vendor's internal rules.
- **Tool / Mechanism:** Vector Database (Supabase pgvector).
- **Input / Output:** Vendor Name + Category -> Extracted corporate policy text (e.g., IndiGo's Conditions of Carriage regarding refunds).²³
- **Decision Logic:** Deterministic semantic search.

6. Regulation Retrieval

- **Trigger & Objective:** Triggered parallel to step 5. The objective is to find the overriding government law.
- **Tool / Mechanism:** Vector Database (Supabase pgvector).
- **Input / Output:** Category -> Statutory text (e.g., DGCA CAR Section 3 or CCPA E-commerce rules).⁷
- **Confidence Handling:** Pre-loading a vector database ensures extremely high confidence and prevents the LLM from hallucinating non-existent laws.

7. Comparative Reasoning (The Core Engine)

- **Trigger & Objective:** Triggered when both policies and regulations are retrieved. The objective is to determine if the vendor violated their own policy or the law.
- **Tool / Mechanism:** High-tier LLM (e.g., GPT-4o) with strict system prompts enforcing logical deduction.
- **Input / Output:** Extracted Facts + Vendor Policy + Government Regulation -> Rationale for claim validity.
- **Decision Logic:** Probabilistic reasoning. The LLM must calculate timelines (e.g., determining if a flight was canceled exactly 12 hours before departure and matching that to the DGCA ₹10,000 compensation tier).⁶ If the reasoning concludes the user has no valid claim, execution routes to a polite rejection message explaining why.

Phase 3: Execution

8. Escalation Route & Contact Discovery

- **Trigger & Objective:** Triggered by a valid claim. The objective is to find where to send the complaint.
- **Tool / Mechanism:** Pre-loaded Nodal Officer Database query.¹¹
- **Input / Output:** Vendor Name -> Officer Name, Designation, Email address.
- **Mocking Strategy:** For the hackathon MVP, live web scraping is too fragile. This step should be a deterministic lookup against a hardcoded JSON directory containing details

for top Indian companies to maintain total credibility without integration risk.

9. Draft Generation

- **Trigger & Objective:** Triggered by contact discovery. The objective is to create the final communication.
- **Tool / Mechanism:** LLM (Drafting prompt).
- **Input / Output:** All previous context -> Formatted, professional email.
- **Guardrails:** The prompt must strictly enforce a polite, firm tone and mandate the explicit citation of the specific clauses extracted in Step 7.

10. Evidence Pack Generation & Human Approval

- **Trigger & Objective:** The final assembly of the artifact.
- **Tool / Mechanism:** PDF Generation Library (e.g., React-pdf).
- **Input / Output:** Summary + Draft + Images -> Downloadable PDF.
- **Human Approval Point:** The workflow halts here. The user must review the draft. An optional integration could offer a "Send via Email" button, but for a hackathon, simply providing a "Copy to Clipboard" or "Download PDF" function is significantly safer and avoids email API spam filters.

SECTION 8: Technical architecture

Designing an architecture for four beginner developers constrained to an eight-day sprint requires extreme pragmatism. Utilizing complex developer frameworks like LangGraph—while industry standard for code-first multi-agent systems—will likely result in spaghetti logic, state management nightmares, and unfinished code.³⁵ A visual, low-code orchestrator combined with a modern frontend framework provides the optimal balance of development velocity, capability, and visual appeal.

Recommended Architecture: The Visual Orchestrator Stack

This stack segregates the UI presentation from the complex agent logic, allowing parallel development streams.

- **Frontend:** Next.js (React) deployed on Vercel. Utilizing Shadcn UI and Tailwind CSS components enables the rapid assembly of a highly polished, professional interface without writing custom CSS. Next.js handles the client-side state and provides API routes to securely communicate with the backend.
- **Backend / Orchestration:** n8n (Cloud hosted). n8n provides a visual node-based editor that perfectly manages data state, handles API rate limits, and allows beginners to visually debug the agent's flow by watching data move between nodes in real-time.³⁷ It excels at linear and branched workflows and easily integrates with LLMs.
- **Database / Vector Storage:** Supabase. Offers PostgreSQL for storing user sessions, object storage for uploaded receipt images, and built-in pgvector capabilities for the

policy Retrieval-Augmented Generation (RAG) pipeline.

- **OCR Vision:** AWS Textract. Built specifically to parse invoices and receipts, outputting highly structured JSON (e.g., isolating `VENDOR_NAME` and `INVOICE_RECEIPT_DATE`) out of the box, which is far more reliable than feeding a raw image to a generic vision model.²⁰
- **LLM Provider:** OpenAI API. Use gpt-4o-mini for the fast, cheap extraction and classification nodes, reserving gpt-4o strictly for the final comparative evaluation and drafting steps to ensure high-quality logical reasoning.

Fallback Architecture: The Python Monolith

If the team struggles heavily with React or asynchronous webhooks, a simpler but less visually impressive architecture can be deployed.

- **Frontend & Backend:** Streamlit or Gradio (Python).
- **Orchestration:** LangChain / LangGraph.
- **Why it is a fallback:** While LangGraph allows for highly sophisticated, stateful agent looping³⁵, it requires a strong grasp of Python decorators, directed graphs, and state typing. Debugging a broken LangGraph loop on day seven of a hackathon poses a severe risk to demo stability. Furthermore, Streamlit applications, while fast to build, often look like generic data science dashboards rather than polished consumer products, potentially harming the "Innovation and Creativity" judging criteria.

Architectural Tradeoffs & Scope Boundaries

- **Safe for beginners:** Building n8n workflows, implementing Shadcn UI components, and utilizing Supabase for basic data storage.
- **Fragile / Overkill:** Fine-tuning local open-source models, implementing real-time web scraping to find nodal officers, or building custom PDF parsing algorithms from scratch.
- **Hackathon Shortcut (MVP Only):** Hardcoding a JSON dictionary of the top 10 company nodal officers¹¹ and pre-loading specific company policies into Supabase manually.
- **Production-Grade Requirements:** A real product would require dynamic web scraping APIs to continually update nodal officer directories⁴⁰, robust observability tools (like Braintrust or LangSmith) to monitor LLM drift and tool failure rates⁴¹, and enterprise-grade authentication.

SECTION 9: Tools, APIs, and data sources

The execution plan relies on integrating highly stable, well-documented tools that offer free tiers suitable for a hackathon environment.

1. Workflow Orchestration:

- **Tool:** n8n (Cloud or self-hosted via Docker).
- **Utility:** Acts as the central nervous system. It triggers on a webhook from the Next.js frontend, routes the data to AWS Textract, passes the results to OpenAI, queries

Supabase, and returns the final draft.

- **Integration & Limitations:** Highly visual and beginner-friendly.⁴³ The cloud trial is sufficient for the hackathon. It is extremely safe and prevents the need to write complex asynchronous polling logic in JavaScript.

2. Document Parsing (OCR):

- **Tool:** Amazon Textract (Analyze Expense API).
- **Utility:** Automatically extracts text, handwriting, and structured data from scanned documents. It explicitly supports standard fields crucial for this product, such as INVOICE_RECEIPT_DATE, VENDOR_NAME, and TOTAL.²⁰
- **Integration & Limitations:** Requires an AWS account and basic IAM configuration, which can be slightly complex for absolute beginners, but the out-of-the-box accuracy on Indian receipts is unmatched. It has a generous free tier for the first 1,000 pages.³³

3. LLM Intelligence:

- **Tool:** OpenAI API (gpt-4o and gpt-4o-mini).
- **Utility:** The cognitive engine for entity extraction, classification, and drafting.
- **Integration & Limitations:** Trivial to integrate via standard API calls or n8n's native nodes. Costs pennies per run. Requires strict system prompting to prevent hallucination.

4. Storage and Vector Retrieval:

- **Tool:** Supabase.
- **Utility:** Provides a generous free tier. The team will use the Postgres database with the pgvector extension to store chunked embeddings of consumer laws and company policies. This allows the agent to perform semantic searches to retrieve the exact legal clause relevant to a specific complaint.

5. Official Data Sources (To be pre-loaded into Supabase):

- **Aviation Regulations:** Directorate General of Civil Aviation (DGCA) Civil Aviation Requirements (CAR) Section 3, Series M, Part II and IV. These documents dictate the exact compensation rules and timelines for delayed and canceled flights in India.⁷
- **E-commerce Regulations:** The Consumer Protection (E-Commerce) Rules, 2020, and the CCPA Guidelines for Prevention and Regulation of Dark Patterns, 2023. These cover unfair trade practices, cancellation charges, and mandatory refund parameters.⁹
- **Company Policies:** Manually scrape and structure the publicly available return policies of Flipkart²² and the Conditions of Carriage for IndiGo Airlines.²³

SECTION 10: Legal, privacy, and trust considerations

When designing a product that interprets policy and generates escalation documents, the system enters a high-risk regulatory environment. While this is a hackathon prototype,

demonstrating an understanding of these risks will highly impress the judges.

Unauthorized Practice of Law (UPL): The most critical product risk is the system being perceived as providing unauthorized legal advice. The FTC's 2024 enforcement action against DoNotPay serves as a stark industry warning; marketing an AI as a "robot lawyer" or claiming it can replace human legal professionals invites intense regulatory scrutiny and consumer harm if the AI fails.²⁹

- **Hackathon Guardrails:** The product must be strictly framed as an administrative drafting assistant. The user interface must feature a prominent, unskippable disclaimer: *"This tool provides administrative drafting assistance based on public corporate policies. It does not constitute legal advice, nor does it establish an attorney-client relationship."*
- **Production Realism:** In a real-world commercial scenario, the output would either need to be reviewed by a human professional or the software's functionality must be strictly limited to customer service corporate escalations, explicitly preventing the generation of formal court affidavits or legally binding contracts.³¹

Privacy and Data Protection (DPDP Act 2023): The Digital Personal Data Protection (DPDP) Act of 2023 radically shifted India's data landscape. It mandates explicit, free, specific, and informed consent for data processing, and requires organizations to implement robust grievance redressal mechanisms of their own.⁴⁵ Flight tickets and e-commerce invoices contain highly sensitive Personally Identifiable Information (PII) such as full names, physical addresses, phone numbers, and partial payment details.

- **Hackathon Guardrails:** Implement a mock "Consent to Process Data" checkbox during the onboarding flow. For the MVP, configure the database to automatically purge all uploaded documents and extracted PII upon the termination of the user's session. Do not save user data persistently.
- **Production Realism:** A production system would require rigorous data retention schedules, automated PII masking before data is sent to external LLM APIs, the appointment of a Data Protection Officer, and verifiable mechanisms for users to exercise their right to erasure.⁴⁷

Hallucination and Output Risks:

If the LLM hallucinates a consumer protection law (e.g., falsely claiming a consumer is owed ₹50,000 for a one-hour flight delay), it destroys the user's credibility when they send the escalation email.

- **Mitigation:** This risk is managed architecturally by restricting the LLM's analytical processing strictly to the text retrieved from the vector database (the RAG pipeline) and utilizing prompt engineering that forces the LLM to explicitly cite the specific section of the policy in its draft, refusing to answer if a citation cannot be found.

SECTION 11: UX and product flow

The user experience must balance extreme simplicity for the consumer with the deliberate, visual exposure of the system's underlying intelligence. A generic chatbot interface (a simple text box and a blinking cursor) will fail to impress the judges; the interface must feel like an autonomous command center.

The Ideal Hackathon UI Flow:

1. **Onboarding Screen:** A clean, high-contrast landing page stating the value proposition. A simple, large dropdown asks: "What did you experience an issue with?" (Options: Flight Disruption, E-commerce Order).
2. **Input & Upload Screen:** A modern split-screen interface. On the left, a large drag-and-drop zone for uploading evidence (receipts, tickets). On the right, a text box stating: "Briefly describe what went wrong."
3. **The "Agent Activity" Screen (The Wow Factor):** Once the user clicks "Analyze Claim", the screen transitions. Instead of a standard loading bar, a terminal-like console component (utilizing Shadcn UI) appears, streaming the agent's step-by-step cognitive process via Server-Sent Events (SSE) from the n8n backend:
 - Initializing OCR extraction pipeline...
 - [Agent] Identified Vendor: IndiGo Airlines. PNR: XYZ123.
 - [Agent] Querying DGCA CAR Section 3 database for cancellation compensation rules...
 - [Agent] Rule identified: Cancellation < 24 hrs warrants full refund + up to ₹10,000 compensation.
 - [Agent] Evaluating vendor policy compliance...
 - [Agent] Drafting escalation notice to Nodal Officer (nodalofficer@goindigo.in)... This visibility is crucial for Agentic UI design patterns.²¹ It visually proves to the judges that complex, multi-step, tool-calling logic is occurring autonomously.
4. **Review Screen:** The generated escalation pack is presented. The screen displays the extracted facts in a clean summary table, the cited policies in an expandable accordion view, and a large, editable text box containing the drafted email.
5. **Action Screen:** Prominent buttons to "Copy to Clipboard" or "Download Evidence PDF".

UI Patterns to Avoid:

Avoid standard conversational chat bubbles. Do not present the agent as an avatar that types out responses character by character. The system should feel like a deterministic processing engine, not a conversational companion.

SECTION 12: MVP scope

Given the highly constrained eight-day timeframe and a team of beginner developers, ruthless scoping is required. The goal is to ensure a completely stable, visually impressive

demonstration on April 1, 2026. The scope must prioritize the illusion of vast capability while hardcoding the boundaries to prevent failure.

Must-Have (The Core MVP):

- Next.js frontend with functional file upload and text input components.
- n8n webhook integration successfully receiving and returning JSON payloads.
- AWS Textract integration capable of pulling text from a clean invoice PDF.
- A static, pre-loaded vector database in Supabase containing rules for exactly two verticals: IndiGo Airlines (Aviation) and Flipkart (E-commerce).
- The "Agent Activity" visual terminal log displaying the processing steps on the frontend.
- A final generated email draft that successfully cites a specific policy retrieved from the database.

Should-Have:

- Automatic identification of the Nodal Officer's email address from a hardcoded JSON directory.¹¹
- Basic error handling (e.g., if the user uploads a blank image, the UI displays an error rather than crashing).

Nice-to-Have (Attempt only if ahead of schedule on Day 6):

- PDF generation of the final summary using a library like react-pdf.
- Conversational fallback (e.g., if the OCR fails to read the date, the agent prompts the user: "I couldn't read the transaction date, could you type it in?").

Explicitly Cut (Do not attempt under any circumstances):

- User authentication or login systems (creates unnecessary friction for a 3-minute demo).
- Live web scraping of real-time company policies (too fragile, high risk of IP blocking or timeout during the demo).⁴⁰
- Actual email transmission via SMTP or APIs like Resend (high risk of spam filters blocking the demo, and unnecessary for proving the concept).

Verticals for MVP Selection:

1. **Aviation (IndiGo):** Showcases the agent's ability to handle complex regulatory math. The DGCA compensation tiers require the agent to calculate the time difference between the cancellation notice and the departure time, proving advanced reasoning.⁶
2. **E-commerce (Flipkart):** Showcases conflict resolution between a platform's restrictive internal policy (e.g., Open Box Delivery no-return rules) and the user's overriding legal rights under the Consumer Protection Act.¹⁹

SECTION 13: Demo strategy

A flawless three-minute live demonstration is the single most important deliverable for hackathon success. The demo must be tightly scripted, theatrical, and utilize pre-prepared, pristine evidence files to entirely eliminate OCR unpredictability. Do not attempt a live, unscripted query during the finale.

Exact Demo Sequence:

1. **The Hook (0:00 - 0:30):** The presenter confidently states the problem. *"Last month, my flight was canceled 12 hours before departure. The airline sent me an automated email offering a travel voucher. But under Indian law, I am entitled to ₹10,000 in cash. Finding that rule and figuring out who to email took me 4 hours. Watch our autonomous agent do it in 30 seconds."*
2. **The Input (0:30 - 1:00):** The presenter uploads a perfectly formatted, pre-tested mock PDF of an IndiGo flight ticket. In the text box, they type: *"My flight was canceled at the last minute and customer support refused a cash refund."*
3. **The Wow Moment (1:00 - 2:00):** The presenter clicks 'Analyze Claim'. The audience watches the Agent Activity terminal log stream the backend processes. The presenter narrates over the action: *"Notice how it's not just generating text. It autonomously extracts the PNR via AWS Textract, recognizes the 12-hour cancellation window, queries our vector database of DGCA regulations, and calculates the exact financial compensation mandated by the government."*
4. **The Payoff (2:00 - 2:30):** The final draft appears on screen. The presenter pauses and highlights the specific paragraph where the LLM cited *DGCA CAR Section 3, Series M, Part IV*⁸, and points out that the email is pre-addressed to the correct Nodal Officer (nodalofficer@goindigo.in).²³
5. **The Conclusion (2:30 - 3:00):** The presenter clicks "Download Evidence Pack", showing the clean, formatted PDF ready for the consumer to use. *"We built a copilot that turns policy opacity into consumer power. Thank you."*

Error Prevention & Fallbacks:

- Always pre-test the demo on the specific Wi-Fi network of the Microsoft Office in Noida before the finale, as corporate firewalls occasionally block non-standard API ports or webhooks.
- **Ultimate Backup Plan:** If the venue Wi-Fi fails entirely, or if the OpenAI API experiences an outage, have a locally hosted, high-definition video recording of the entire seamless workflow ready to play. The presenter can narrate over the video identically to the live demo.

SECTION 14: PPT strategy

The 5-6 slide presentation must be visually minimalist and map directly to the hackathon evaluation criteria: Innovation (25%), Technical Execution (25%), Problem-Solution Fit (20%), and Impact (20%).

- **Slide 1: Title & Vision.** (Maps to Problem-Solution Fit)
 - *Title:* Consumer Claims Agent
 - *Message:* Transforming consumer frustration into structured resolution.
 - *Visual:* Clean logo, one-line concept.
- **Slide 2: The Scale of the Problem.** (Maps to Impact)
 - *Title:* The Bureaucracy Barrier
 - *Message:* Indian consumers are abandoning millions in valid refunds due to policy opacity and administrative friction.
 - *Proof:* Cite the 5.15 lakh pending court cases and the NCH's 1.55 lakh monthly calls.¹
- **Slide 3: The Solution.** (Maps to Innovation)
 - *Title:* Autonomous Escalation
 - *Message:* We built an agentic workflow, not a chatbot. It does the research so the user doesn't have to.
 - *Visual:* A side-by-side comparison chart. "Standard Chatbot" = User must know the law. "Our Agent" = System retrieves the law automatically.
- **Slide 4: Under the Hood.** (Maps to Technical Execution)
 - *Title:* Agentic Architecture
 - *Message:* A robust, scalable, "Plan-and-Execute" architecture.
 - *Visual:* A clean architectural flow diagram showing the Next.js frontend, the n8n orchestrator routing logic, AWS Textract, and the Supabase RAG database.
- **Slide 5: Legal & Ethical Guardrails.** (Maps to Impact & Scalability)
 - *Title:* Built for Safety
 - *Message:* Engineered for compliance in 2026.
 - *Proof:* Highlight DPDP Act 2023 compliance (automatic data purging) and UPL avoidance (strict administrative copilot framing).

30-Second Pitch:

"Every day, thousands of Indian consumers abandon valid refunds because fighting corporate bureaucracy is exhausting. We built the Consumer Claims Agent, an autonomous workflow that ingests your unstructured receipts, cross-references them against official government regulations like the DGCA charter, and instantly generates a legally sound escalation package to get your money back, without you needing to read a single page of legal jargon."

90-Second Pitch:

(Expand the 30-second pitch by detailing the architecture): "...To achieve this, we didn't just build a ChatGPT wrapper. We built a true 'Plan-and-Execute' agent using n8n. When a user uploads an image, our orchestrator routes it to AWS Textract for deterministic data extraction. It then passes those facts to a vector database containing thousands of pages of corporate policy and consumer law. Finally, an LLM evaluates the intersection of those facts and laws to draft a highly specific, cited escalation notice targeted directly at the company's nodal officer."

Anticipated Judge Questions & Strong Answers:

1. Q: *Why not just use ChatGPT?*
 - A: "ChatGPT requires the user to know the law to prompt it correctly, and it notoriously hallucinates legal precedents. Our system is built on a RAG pipeline hooked into deterministic databases. We force the LLM to only cite the policies we have verified, and our orchestration layer handles the tool-calling for OCR automatically."
2. Q: *Is this legally safe? Aren't you practicing law?*
 - A: "No. We specifically designed and framed this as an administrative copilot. We are structuring customer service complaints based on public corporate policies, not filing court affidavits or acting as legal counsel. It is an escalation tool, not a litigation tool."
3. Q: *How is this different from customer support automation?*
 - A: "Enterprise customer support bots are built to protect the company's margins and deflect tickets. The Consumer Claims Agent is built symmetrically to protect the consumer's time and enforce their rights. It is an offensive tool for the consumer, not a defensive tool for the corporation."
4. Q: *What happens if the OCR fails to read the receipt?*
 - A: "Our n8n workflow has conditional logic. If the extraction confidence drops below 80%, the agent halts the autonomous flow and elegantly falls back to prompting the user to manually enter their PNR or invoice number."
5. Q: *How do you handle data privacy with these sensitive receipts?*
 - A: "In alignment with the DPDP Act 2023, we practice strict data minimization. For this MVP, session data and uploaded images are processed in-memory and purged immediately after the draft is generated. We do not store consumer PII."

SECTION 15: Implementation plan

Executing this complex architecture with a four-person beginner team between March 24 and April 1, 2026, requires parallel workstreams and strict daily milestones. The team must avoid "blocking" dependencies; the frontend must be built simultaneously with the backend using mock data payloads.

Team Role Allocation:

- **Dev 1 (Frontend Lead):** Owns the Next.js UI, Shadcn component implementation, state management, and the visual "Agent Activity" terminal log.
- **Dev 2 (Orchestration Lead):** Owns the n8n workflows, prompt engineering, conditional routing logic, and OpenAI API connections.
- **Dev 3 (Data & Integrations):** Owns Supabase setup, chunking and vectorizing the legal PDFs, and configuring AWS Textract.
- **Dev 4 (Product/QA):** Owns the mock data generation (creating the perfect test PDFs), PPT design, testing edge cases, hardcoding the Nodal Officer JSON, and scripting the live demo.

Day-by-Day Execution Plan:

- **Day 1 (Mar 24): Architecture & Infrastructure.** Initialize the GitHub repository. Set up Supabase, Vercel, and n8n cloud instances. Agree on the exact JSON schema that the frontend will send to the webhook.
- **Day 2 (Mar 25): Data Gathering & Vectorization.** Dev 3 manually scrapes and vectorizes the DGCA passenger charter and Flipkart return policies into Supabase. Dev 4 creates the "Golden Path" mock PDFs for IndiGo and Flipkart.
- **Day 3 (Mar 26): Core Workflow Construction.** Dev 2 builds the linear n8n pipeline: Webhook -> AWS Textract -> LLM Entity Extraction. Test this sub-graph using Postman.
- **Day 4 (Mar 27): UI Foundations.** Dev 1 builds the onboarding and upload screens in Next.js. Dev 2 adds the Supabase vector search and final LLM drafting nodes to the n8n workflow.
- **Day 5 (Mar 28): Integration Day.** Connect the Next.js frontend to the n8n webhook. Ensure the file payload passes correctly and the final draft string is returned to the UI.
- **Day 6 (Mar 29): The Agent Log (The Wow Factor).** Dev 1 and Dev 2 collaborate to send intermediate progress updates from n8n to the Next.js UI, populating the visual terminal log.
- **Day 7 (Mar 30): QA and Bug Triage.** Stop building new features. Test the end-to-end flow exclusively with the two chosen demo scenarios. Fix OCR formatting bugs and refine the LLM prompts to ensure the tone is perfect.
- **Day 8 (Mar 31): Demo Lock.** Code freeze. Record the high-definition backup video. Rehearse the 3-minute script. Finalize the PPT slides.
- **April 1:** Final Submission.

Repository and Testing Strategy:

Utilize a simple Git branching strategy: a main branch for the stable demo code, and individual feature branches (e.g., ui-upload, n8n-integration). Avoid writing automated unit tests; in an 8-day hackathon, time is better spent on manual integration testing against the two specific "Golden Paths." During the final 48-hour bug triage, if a bug cannot be fixed within 2 hours, hardcode a workaround for the demo scenario.

SECTION 16: Technical depth for judges

When interacting with the judging panel or mentors, the team must articulate their architectural choices clearly, conveying deep technical maturity while avoiding empty industry buzzwords.

Explaining the Agent Architecture: Do not simply call it an "AI." Describe the system as utilizing a "Plan-and-Execute" agentic design pattern.³² Explain that the orchestration layer (n8n) acts as a deterministic control mechanism, safely bounding the probabilistic nature of the LLM.

Explaining Tool Use and RAG:

Emphasize the strict separation of deterministic and probabilistic functions. State clearly: "We do not use LLMs for tasks they are inherently bad at. We use AWS Textract for deterministic data extraction, and we use vector database queries for deterministic policy retrieval. We only use the LLM for its core strength: cognitive evaluation and text synthesis. This hybrid RAG (Retrieval-Augmented Generation) approach mathematically bounds the context window, guaranteeing accuracy and virtually eliminating legal hallucination."

Explaining Multi-Step Reasoning: Walk the judges through the ReAct (Reason + Act) loop.¹⁷

Explain that the system doesn't just read the prompt and write an email. It reads the image, *reasons* that it needs a vendor policy, *acts* by querying the database, *reasons* about the difference between the policy and the law, and finally *acts* by generating the draft. Point to the UI's Agent Activity log as visual proof of this execution graph.

SECTION 17: Risk register

The following practical risk table outlines the primary threats to hackathon success and their corresponding mitigation strategies.

Risk	Likelihood	Impact	Prevention Strategy	Fallback Plan
OCR Failure on Live Receipts	High	Critical (Breaks the entire pipeline)	Use high-quality, pre-tested mock PDFs for the live demo. Do not use random images.	Build the UI to allow manual text entry of the PNR/Invoice number if the image upload fails.
Vector DB / Legal Hallucination	Medium	High (Loss of credibility with judges)	Keep vector chunk sizes small. Use strict prompt engineering commanding the LLM to <i>only</i> use provided context.	Hardcode the exact legal policy text into a hidden prompt field exclusively for the demo path.

API Timeout / Wi-Fi Failure during Demo	Medium	Critical	Optimize prompt sizes to reduce latency. Verify Microsoft Noida venue network ports.	Immediately switch to the pre-recorded, high-definition seamless video of the workflow.
UPL / Legal Scrutiny	Low (for hackathon)	High	Explicitly brand and pitch the product as an "Administrative Copilot," not an AI lawyer.	Point directly to the prominent liability disclaimers built into the UI design.
Over-scoped MVP / Unfinished Code	High	Critical	Strictly limit the database and logic to only 2 verticals (Aviation/Ecom). Cut authentication entirely.	Submit a partially working prototype with a highly polished frontend facade that simulates backend responses.

SECTION 18: Sample outputs

The following scenarios illustrate the exact data flow and expected outputs for the two primary demonstration paths.

Scenario 1: Aviation Disruption (The Golden Demo Candidate)

- **User Input:** "IndiGo canceled my flight from Kolkata to Bangalore 12 hours before it was supposed to leave. They just sent me an email offering a travel voucher. I want my money back in cash."
- **Uploaded Evidence:** A mock PDF of an IndiGo E-Ticket.⁴⁹
- **Extracted Facts (via OCR & LLM):** Vendor: IndiGo Airlines. Route: CCU to BLR. Issue: Flight Cancellation. Timeline: < 24 hours notice.
- **Identified Policy/Regulation:** *DGCA CAR Section 3, Series M, Part IV.* (The system retrieves the specific clause mandating that passengers notified of a cancellation less than

24 hours before departure are entitled to a full refund and compensation up to ₹10,000).⁶

- **Escalation Route:** Nodal Officer, IndiGo Airlines (nodalofficer@goindigo.in).²³
- **Sample Draft Excerpt:** *"Dear Nodal Officer, I am writing to formally escalate the cancellation of PNR XYZ123. I was notified of this cancellation exactly 12 hours prior to departure. Under DGCA CAR Section 3, Series M, Part IV, passengers subjected to cancellations with less than 24 hours notice are legally entitled to a full refund to the original payment source and monetary compensation. I formally decline the offer of a travel voucher and request the processing of my statutory cash refund and compensation within the mandated 7-day period..."*

Scenario 2: E-Commerce Damaged Goods

- **User Input:** "I bought a blender on Flipkart. The box looked fine when the delivery guy dropped it off, but when I opened it, the glass jar was completely shattered. The app says it's not returnable because it was an open box delivery."
- **Uploaded Evidence:** A mock PDF Flipkart invoice.
- **Extracted Facts:** Vendor: Flipkart. Issue: Damaged electronics upon opening. Policy applied by vendor: Open Box Delivery refusal.
- **Identified Policy/Regulation:** *Consumer Protection (E-Commerce) Rules, 2020.* (The system retrieves the clause stating platforms cannot refuse to take back defective goods or impose unfair cancellation policies, superseding internal platform rules for concealed damage).⁹
- **Sample Draft Excerpt:** *"Dear Grievance Officer, I am writing regarding Order #12345 containing a severely damaged product. While your automated support system rejected the return citing an Open Box Delivery policy, the shattered glass was internal and not visible upon external inspection at the door. Under the Consumer Protection (E-Commerce) Rules, 2020, platforms are statutorily obligated to facilitate returns for defective goods, regardless of internal open-box policies when the defect is concealed. Please arrange a reverse pickup..."*

SECTION 19: Final recommendation

The Consumer Claims Agent is an exceptionally strong, top-tier hackathon concept. It perfectly threads the needle between technical innovation, strict track alignment, and deep, provable market relevance in India. By focusing on a systemic administrative pain point rather than a trivial consumer novelty, the team demonstrates product maturity.

Summary of Strategic Directives:

- **Best Final Framing:** A "Grievance-Resolution Copilot" that structures administrative escalations, strictly avoiding the "AI Lawyer" label to mitigate UPL risks.
- **Recommended Stack:** React (Next.js) + Shadcn UI + n8n + Supabase (pgvector) + AWS Textract + OpenAI API.
- **Recommended MVP Scope:** Strictly limited to IndiGo flight cancellations and Flipkart

damaged goods, prioritizing flawless demo stability over broad vertical capability.

- **Recommended Demo Scenario:** The IndiGo flight cancellation, as it brilliantly showcases the agent's ability to execute complex regulatory math regarding DGCA compensation tiers.
- **Things to Avoid:** Do not build a chatbot interface. Do not attempt live web scraping. Do not build authentication.

Top 10 Actions to Take Immediately (Next 72 Hours):

1. Initialize the GitHub repository and configure the Next.js foundation.
2. Register for a Supabase account and instantiate the Postgres database with the vector extension.
3. Download the official DGCA Passenger Charter PDF and the Flipkart Return Policy text.
4. Chunk and vectorize these two specific documents into the Supabase database.
5. Create two pristine, highly legible mock PDFs (one IndiGo ticket, one Flipkart invoice) to serve as the golden test data.
6. Connect the n8n instance to AWS Textract and verify clean data extraction using the mock PDFs.
7. Draft the strict system prompts for the LLM evaluation and drafting nodes.
8. Build the Next.js UI shell, focusing heavily on the visual "Agent Activity" terminal log component.
9. Connect the Next.js frontend to the n8n webhook and ensure data flows cleanly back and forth.
10. Begin end-to-end integration testing of the IndiGo scenario.

Appendix A: Condensed Build Brief

Project: Consumer Claims Agent (Track 2: Agentic AI)

Objective: Build a highly polished 3-minute demo showing an autonomous agent extracting receipt data, querying legal rules, and drafting an escalation email.

Architecture Flow: Next.js UI (Input) -> n8n Webhook -> AWS Textract (OCR) -> LLM (Fact Extraction) -> Supabase (Policy RAG Search) -> LLM (Evaluation & Drafting) -> Next.js UI (Output & Terminal Log).

Core Constraint: Do not build a conversational chatbot. Build a goal-oriented interface that visually exposes the step-by-step system execution and reasoning loop.

Appendix B: Final MVP Checklist

- [] User Input UI constructed (Text area + Drag-and-drop file upload).
- [] n8n webhook connection successfully established and tested.
- [] AWS Textract OCR extraction working flawlessly on the specific test PDFs.

- [] Supabase vector database populated with the IndiGo and Flipkart policies.
- [] LLM drafting node configured to generate polite emails with strict policy citations.
- [] Next.js UI successfully visualizes the "Agent Activity" processing steps in a terminal view.
- [] Static JSON Nodal Officer contact database linked and functional.
- [] Output screen allows the user to copy the final text to their clipboard.

Appendix C: Final Demo Checklist

- [] Ensure the presenter's laptop is successfully authenticated on the Microsoft Noida venue Wi-Fi.
- [] Have the pristine IndiGo mock PDF saved directly on the desktop for fast access.
- [] Have the exact text prompt pre-written in a notepad file to copy-paste, avoiding live typos.
- [] Open the n8n execution log on a secondary monitor (if available) to verify background processing during the pitch.
- [] Pre-load the high-definition backup video on a media player, ready to switch tabs instantly if the API or internet fails.
- [] Rehearse the 3-minute presentation script multiple times to align perfectly with the system's generation latency.

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